Multivariable Analysis for Event Selection of LHCb data

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June 2026

Abstract

This study details the investigation and isolation of the $B_s^0 \to \psi(2S)K_S^0$ decay in LHCb experimental data. Given the low signal-to-background ratio, a multivariate selection strategy based on Machine Learning was implemented. Simulated data, calibrated via kinematic weighting and the sPlot technique on a B^0 control channel, were utilized alongside background samples in the Upper Sideband of the invariant mass distribution to train a Boosted Decision Tree classifier. After applying a stringent selection on the algorithm's output to suppress the combinatorial background, an extended maximum likelihood fit was performed on the reconstructed invariant mass distribution. The analysis yielded a clear extraction of the B_s^0 signal with a statistical significance of 4.12σ.

1 Introduction

1.1 LHCb detector

The LHCb experiment [1] is a sophisticated particle physics detector located at the Large Hadron Collider (LHC) at CERN. It is specifically designed to perform precision measurements to study the differences between matter and antimatter by investigating CP violation in rare decays of particles containing b (bottom) and c (charm) quarks.

LHCb is a single-arm forward spectrometer optimized to capture particles produced in the forward direction. This is crucial because b-hadrons are typically produced in the forward region with high longitudinal boost due to their momentum. A schematic view of the detector is shown in Figure 1. A right-handed coordinate system is used for description in the image: the z-axis is along the beam and the y-axis is vertical. The negative (positive) z-region of the detector is referred to as upstream (downstream).

The LHCb detector consists of several subsystems with different utilization;

- **Vertex Locator (VELO):** It is placed surrounding the interaction region, just millimeters away from the LHC beams. Composed of silicon pixel sensors, the VELO provides exceptional spatial resolution. It is critical for separating primary collision vertices from secondary vertices of short-lived beauty and charm states.

- **Dipole Magnet:** A large, non-superconducting conventional magnet generating an integrated magnetic field of approximately 4 Tm perpendicular to the horizontal bending plane. It deflects charged particles to enable momentum measurement and particle identification.

- **Tracking system:** The systems include the Trigger Tracker (TT or Tracker TUricensis) which is located upstream of the dipole magnet, and three tracking stations (T1, T2, T3) which are located downstream of the magnet. These components measure the momentum of charged particles by tracking their paths through the magnetic field, which bends the trajectories of charged particles depending on their momentum. The tracking stations also consist of drift tubes filled with gas which generate signal from ionization.

- **RICH detectors:** The RICH detectors distinguish charged hadrons (pions, kaons, and protons) by utilizing the Cherenkov effect, where particles traveling faster than the speed of light in a medium emit Cherenkov radiation. The RICH detectors, using different radiators (aerogel, C4F10, CF4), cover a wide range of momenta and provide high particle identification efficiency. By measuring the angle of Cherenkov radiation and combining it with momentum information, the particle's mass and type can be determined.

- **Calorimeter:** The full calorimeter system consists of an electromagnetic calorimeter (ECAL), a hadronic calorimeter (HCAL), a scintillating pad detector (SPD), and a preshower detector (PS). These devices measure the energy of particles by absorbing them and producing scintillation light, which is proportional to the energy deposited. ECAL is optimized for detecting photons and electrons, while HCAL is designed for hadrons. These measurements are essential for identifying particles and understanding their energy distributions.
- **Muon system:** The muon system comprises multiple layers of muon chambers alternated with iron absorbers. This configuration ensures high-efficiency muon identification, which is essential for analyzing decay processes involving muons, which are the minimum ionization particle (MIP) at LHC.

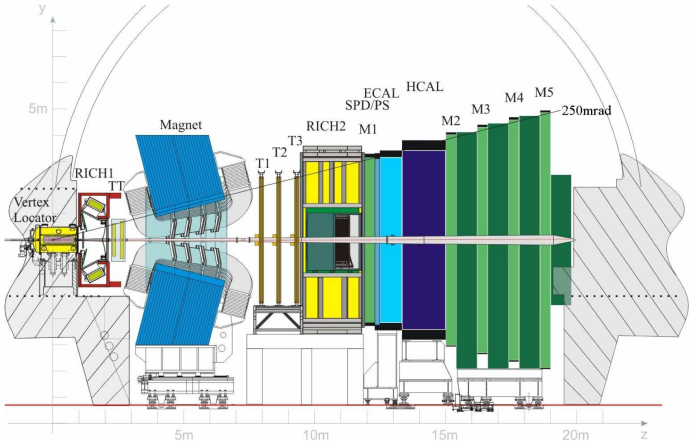


Figure 1: Schematic overview of the LHCb detector and its subsystems.

1.2 B_s^0 meson physics

This report referred to the one of the B_s^0 meson decay (contains a s-quark and a $\bar{b}$-quark) which is flavour changing weak decay ($\bar{b} \to \bar{d}$ transition), $B_s^0 \to \psi(2S)K_s^0$. The $\psi(2S)$ meson consists of a $c\bar{c}$ quark pair and the K_s^0 meson is a superposition between $d\bar{s}$ and $s\bar{d}$ states. The leading order Feynman Diagram is shown in 2.

Nevertheless, these two mesons cannot be observed directly by the detector. This required the reconstruction from their products. The $\psi(2S)$ meson is reconstructed from two oppositely charged muons with invariant mass, $m(\mu^+\mu^-) \simeq m(\psi(2S))$, This decay is preferable because muon tracks are easy to reconstruct and seldom mis-identified due to muon system in LHCb detector. The other $c\bar{c}$ mesons are negligible in the dataset due to applying di-muons

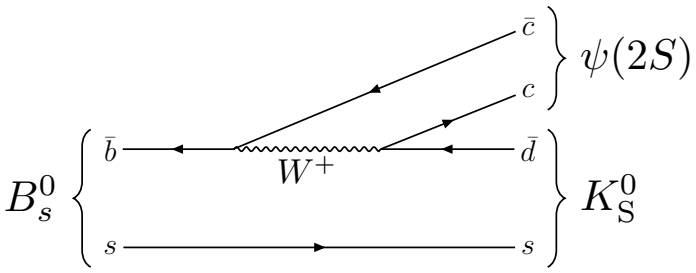


Figure 2: Feynman diagram of hadronic decay of the B_s^0 meson into $\psi(2S)$ and $K_{\rm S}^0$ states

invariant mass selection. The K_s^0 meson is reconstructed from two oppositely charged pions due to the dominant decay of K_s^0 meson[1].

The dataset contains the measured variables of the muon and pion which are the reconstructed candidate from both of mesons. With this dataset, the B_s^0 meson can be reconstructed from the $\psi(2S), K_s^0$ mesons while assuming the true nominal masses of them. the dataset also consists of the combinatorial background and the decay $B^0 \to \psi(2S)K_s^0$ which is kinematically similar to the B_s^0 decay. The invariant mass of the B^0 events shows a sharp peak that is smeared out due to the detector resolution. Combinatorial background results from wrongly combined events that do not originate from the decay of the B^0 or B_s^0, respectively. The invariant mass of the combinatorial background is distributed exponentially. The signal need to be separated from the background by implemented Machine Learning as discussed in Section 2.

2 Data analysis

The primary objective of this analysis is the observation and yield estimation of the B_s^0 meson decay. Since the signal is deeply covered by a dominant combinatorial background, a simple cut-based selection on kinematic variables would be insufficient to isolate the signal peak. Therefore, a multivariate analysis strategy was adopted, and the following steps were followed.

Given the overwhelming background in the real data, distinct signal and background control regions were first defined. We then identified and selected the most highly discriminating features to separate the two categories. Subsequently, a Boosted Decision Tree (BDT) classifier was trained on these features to assign a probability score reflecting the signal-like nature of each event, allowing us to apply an optimized threshold for background rejection. Finally, this classification was applied to the full dataset, and an extended maximum likelihood fit was performed on the resulting invariant mass distribution to extract the signal yield and quantify the statistical significance of the $B_s^0 \to \psi(2S)\, K_S^0$ observation.

[1]The branching ratio is: $\mathfrak{B}(K_s^0 \to \pi^+\pi^-) = (69.20 \pm 0.05)\%$

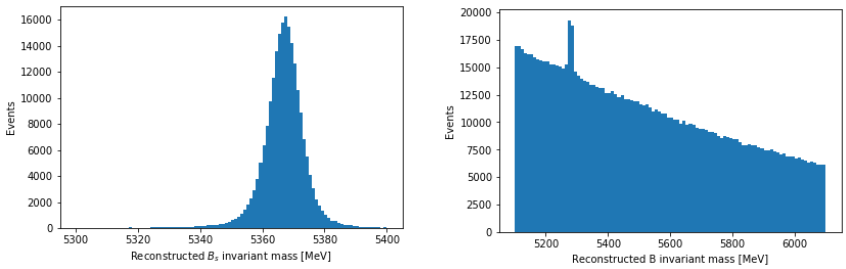


Figure 3: Invaraint mass distribution of B_s^0 simulation (left) and real data (right)

2.1 Training samples

Since the Boosted Decision Tree (BDT) is a supervised learning algorithm, it requires explicitly labeled signal and background samples for training. To represent the pure signal, Monte Carlo (MC) simulations were utilized. However, simulated events often fail to perfectly reproduce the kinematic spectra of real data. To resolve this discrepancy and ensure the MC accurately reflects the real detector conditions, *kinematic weights* were applied to the simulated sample. These weights re-calibrate the simulated distributions to match the observed data, thereby preventing the BDT from artificially learning the non-physical differences between MC and real data.

Conversely, relying on simulations to model the background is generally inadequate due to the inherent complexity and unpredictability of combinatorial processes. Therefore, the background training sample was extracted directly from the real dataset by selecting events from the Upper Sideband (USB) of the invariant mass distribution. The USB, defined as the region with reconstructed masses significantly higher than the nominal B_s^0 mass, is chosen because it is naturally free of physical backgrounds (which typically populate the lower mass regions due to missed particles in partially reconstructed decays) and contains exclusively pure combinatorial background. For this analysis, the signal window was defined between 5333 and 5394 MeV/c^2, and the background was sampled from the region above this upper boundary.

A fundamental challenge in this approach is ensuring that the classifier learns to distinguish true physical signal from background, rather than merely distinguishing simulation artifacts from real data. Since the B_s^0 signal in the data is too obscured by background to validate the MC directly, we exploited the kinematically similar and much more abundant $B^0 \to \psi(2S)K_S^0$ decay as a control channel.

To properly compare the real B^0 data with the B^0 simulations, we employed the *sPlot* statistical technique. Applying these sWeights to the real data effectively tranforms the dataset, statistically subtracting the background contribution and making it much more similar to a pure B^0 signal distributions (Figure 5. This crucial step allowed us to rigorously evaluate the data-simulation agreement and ensure the reliability of our MC approach.

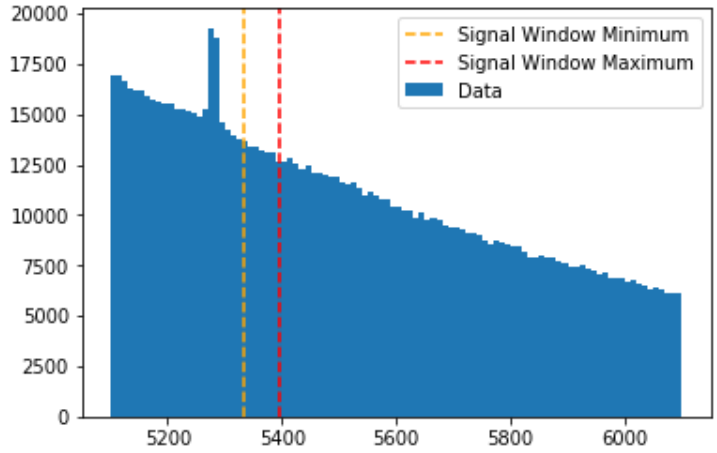


Figure 4: Distribution of the invariant mass from the real dataset, showing the defined signal window (5333 - 5394 MeV/c^2) considered for the analysis.

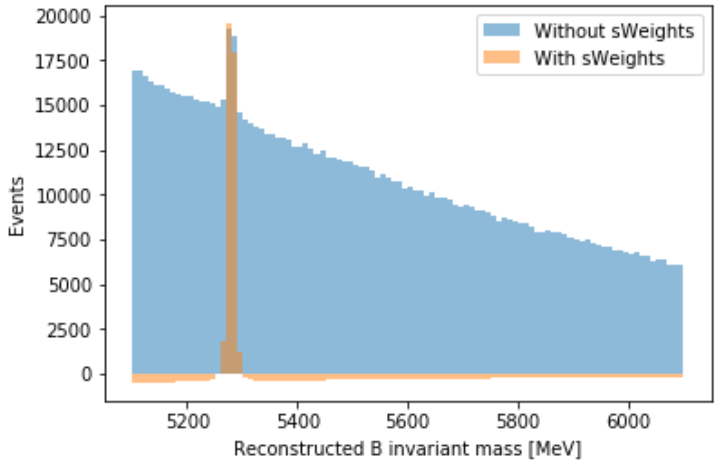


Figure 5: Reconstructed invariant mass distribution of the control channel before (in blue) and after (in orange) the application of sWeights, demonstrating the extraction of the pure signal component.

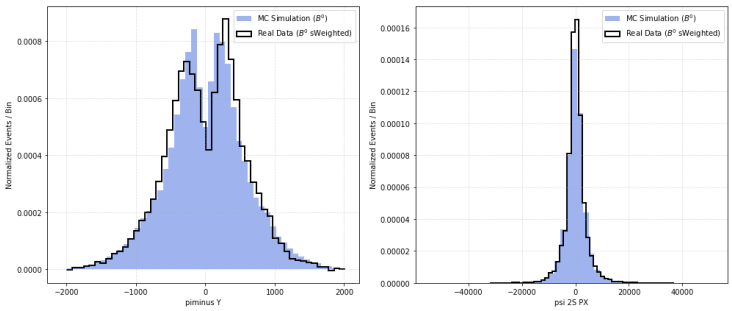


Figure 6: Example of simulated and real data distributions of selected features which satistfy the data-MC agreement criterion chosen.

2.2 Feature selection

Providing the BDT classifier with robust, independent, and physically meaningful variables is crucial to prevent the algorithm from learning non-physical artifacts. Not all available kinematic and topological features are perfectly modeled by the simulation, and not all of them have good discriminating power. Therefore, to ensure a robust classification, the input features were required to satisfy a strict set of logical criteria.

To quantitatively evaluate the similarity and the separation power between different distributions, the Kolmogorov-Smirnov (KS) test was utilized. The KS statistic, d, computes the maximum absolute distance between the empirical cumulative distribution functions (CDFs), $F^1(x)$ and $F^2(x)$, of two given samples:

$$d = \sup_x \left| F^1(x) - F^2(x) \right| \tag{1}$$

A smaller d value indicates a higher similarity between the two distributions, while a larger d value signifies strong separation. Using this metric, the feature selection was performed through the following sequential steps:

1. **Data and Simulation Agreement:** To ensure the BDT learns physical differences of signal from background rather than simulation from real data, the KS distance was computed between the simulated B^0 control channel and the real B^0 data (extracted via sWeights). Only features demonstrating strong data-MC agreement (KS distance < 0.03) were retained. Some examples of the distribution for features which satisfy the agreement were shown in Figure 6.

2. **Separation Power:** Surviving features must effectively discriminate signal from combinatorial noise. Therefore, a second KS test was performed between the B_s^0 MC signal and the real data background, exctracted from the USB, selecting only variables with substantial separation power (KS distance > 0.1). Some examples of features satisfying the separation cut are shown in Figure 7.

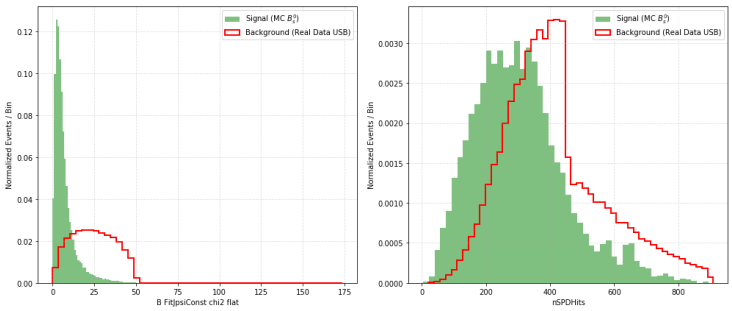


Figure 7: Example of distributions of features which have a lot of separation power between signal and background.

3. **Mass Independence and Uncorrelation:** To avoid using variables which were highly correlated to the B_s^0 invariant mass, all features with a correlation greater than 0.1 with $m_{B_s^0}$ were discarded. Furthermore, to avoid redundant information and optimize training, mutual correlations among the surviving variables were evaluated to iteratively remove highly inter-correlated features. The correlation matrix is shown in Figure 8.

After strictly applying these criteria and incorporating the proper kinematic weights and sWeights in the metric calculations, a final, highly optimized subset of 28 independent features was selected to train the classifier.

2.3 Multivariate Classification and Threshold Optimization

To separate the physical signal from the combinatorial background, the selected variables were used to train a Boosted Decision Tree (BDT) , a multivariate classifier implemented through the XGBoost implementation [2]. A BDT is an ensemble machine learning algorithm that sequentially combines multiple "weak learners" (simple decision trees) to create a highly accurate predictive model. In the boosting process, each subsequent tree is trained to correct the misclassifications made by the previous ones. This iterative approach allows the algorithm to learn complex, non-linear correlations among the input kinematic features and output a probability score indicating how "signal-like" an event is.

Training Strategy and Cross-Validation

The classifier was provided with the best independent features selected in the previous step. However, before the training, it's important to note that the training sample exhibited a significant class imbalance, with background events heavily outnumbering signal events. So, to prevent the algorithm from developing an artificial bias towards the dominant background class, the dataset was reweighted, ensuring that both classes had the same statistical contribution to

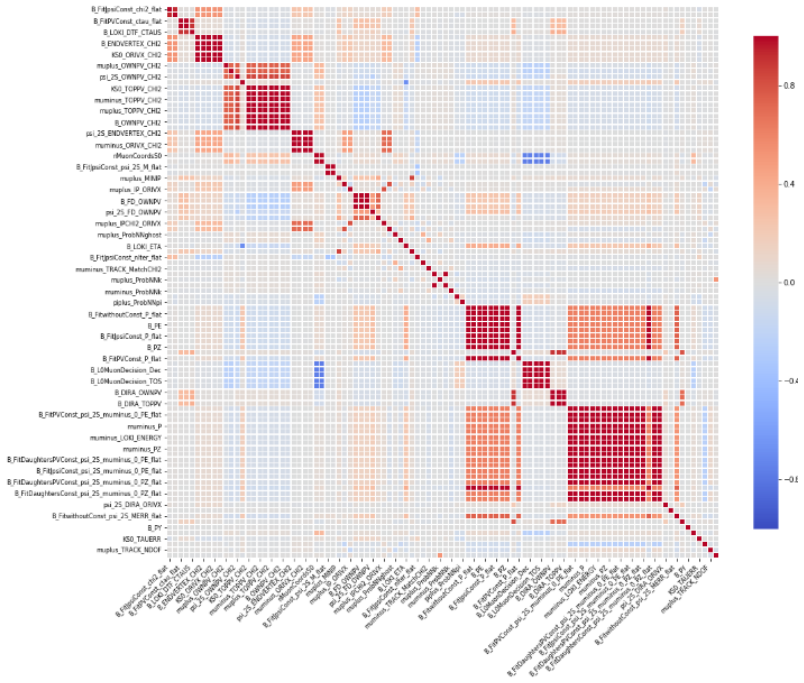


Figure 8: Correleation matrix of all features in the dataset which passed the data-MC agreement and the separation power criteria.

the learning process. Note that the distributions of the response variable were normalized by BDT.

Furthermore, a k-fold cross-validation procedure (with $k = 5$) was implemented to prevent overtraining. The full training sample was randomly partitioned into five equal subsets. The BDT was then trained five independent times, each time using four subsets for training and the remaining subset for validation. This technique allows the model to leverage the entire dataset for both training and validation without introducing statistical bias, maximizing the available statistics while ensuring robust generalization.

The performance of the training is studied using a receiver operating characteristics (ROC) curve. It shows the true positive rate (TPR) as a function of its false positive rate (FPR), corresponding to different cut values on the classifier's output. It can be evaluated with the area under the ROC curve (AUC) which equals 1 for a perfect classifier and 0.5 for a randomly guessing classifier. The ROC curve and the prediction distributions were shown in Figure 9.

From the ROC curve, the AUC score is 0.99 approximately. It means that the curve leans towards the side where TPR > FPR, confirming the good performance of the classifier.

Classification Efficiency and Optimal Cut Selection

Figure 9 also shown the output distribution of BDTs for signal and background. Most signal samples are classified with a high output probability while background samples mostly have a small probability to be signal, which is the desired behaviour of the BDTs. The abscence of many signal samples with low output probability or the other way round also exclude the possibility of overfitting to impact the output. Additionally, the small amount of medium probabilities visualise the separation power of the classifier.

But, to translate this predictive power into a physical selection, the behavior of the classification efficiencies must be analyzed. Figure 10 displays the signal and background efficiencies as a function of the BDT cut value. As the threshold increases, the background efficiency drops significantly faster than the signal efficiency, confirming the excellent discrimination power of the model.

To determine the exact optimal threshold on where to cut the BDT output, the Punzi Figure of Merit (FOM) [3] was utilized. The FOM provides a quantitative mathematical metric to maximize the physical sensitivity of the search, defined as:

$$\text{FOM} = \frac{\epsilon_{sig}}{\frac{5}{2} + \sqrt{N_{bkg}}} \tag{2}$$

where ϵ_{sig} is the signal efficiency and N_{bkg} is the expected number of background events surviving the cut. The signal efficiency (ϵ_{sig}) was calculated as the weighted fraction of simulated signal events that pass the BDT selection, using the provided kinematic weights to ensure physical accuracy. The number of expected background events (N_{bkg}) was estimated directly by counting the number of real data events in the signal region that satisfy the BDT selection criteria.

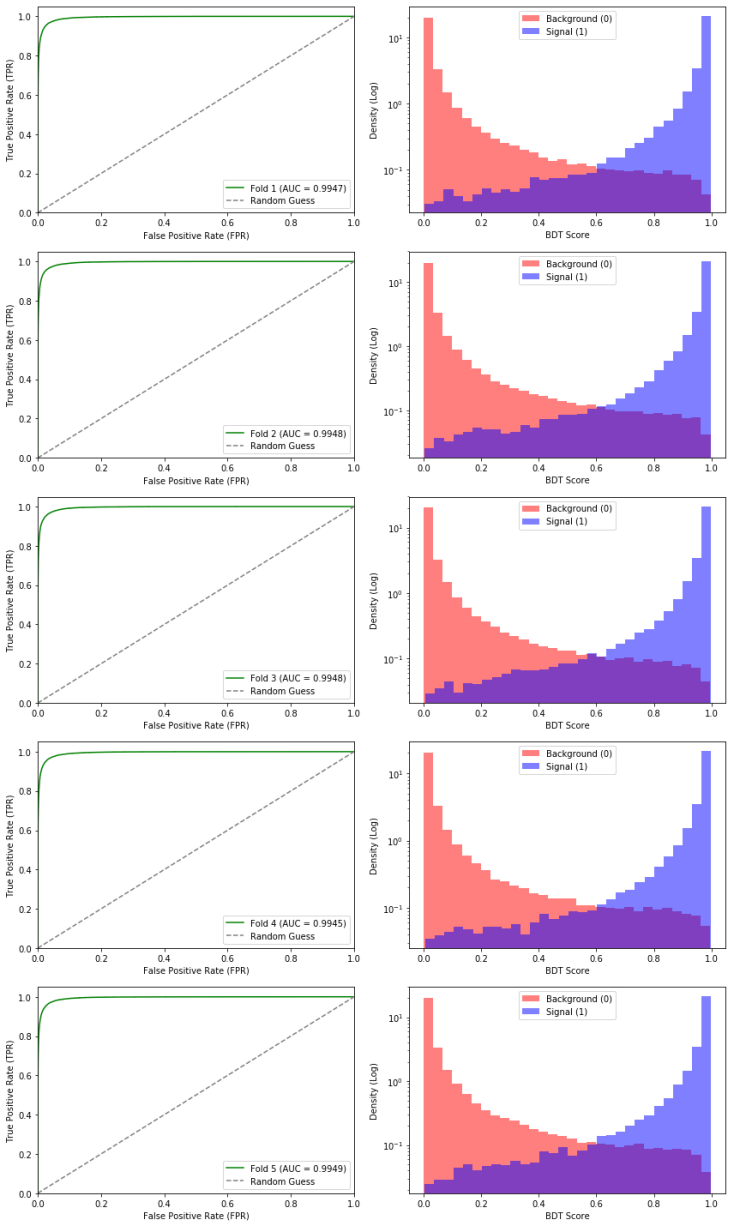


Figure 9: ROC curve with AUC score and the output probabilities of signal (blue) and background (red) for each BDT

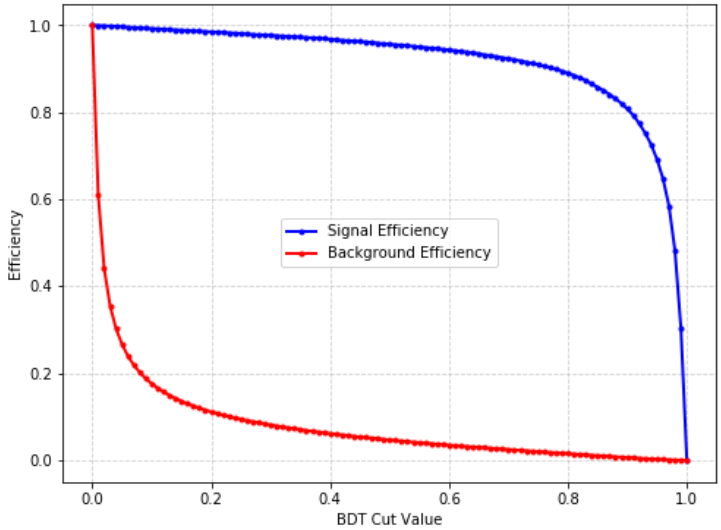


Figure 10: Signal (blue) and background (red) classification efficiencies as a function of the BDT output threshold.

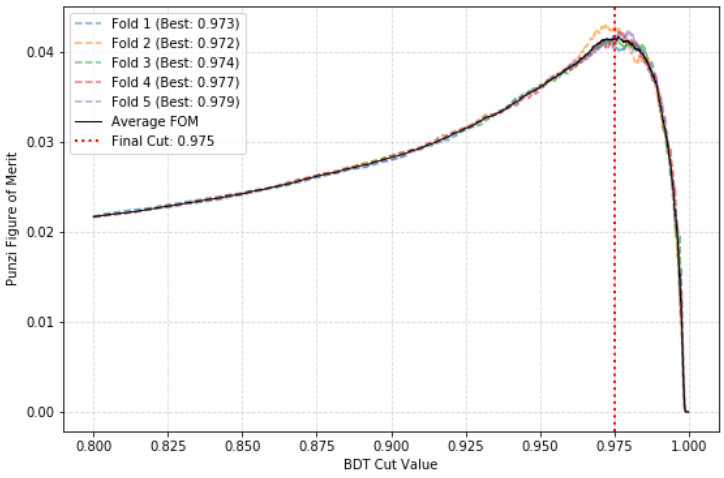


Figure 11: Punzi Figure of Merit evaluated as a function of the BDT threshold cut. The maximum sensitivity is achieved at a cut value of 0.975.

As illustrated in Figure 11, the FOM distribution exhibits a clear peak at a cut value of 0.975. Consequently, this value was adopted as the definitive criteria to separate signal candidates from background noise.

Invariant Mass Distribution After Selection

The classification threshold was ultimately applied to the full unblinded dataset. The resulting invariant mass distribution of the surviving candidates is shown in Figure 12. The stringent BDT cut (> 0.975) successfully eliminated the vast majority of the overwhelming combinatorial background. The final spectrum reveals a highly purified sample, featuring a prominent B^0 control peak and a clearly visible B_s^0 signal peak polluted with a remaining, but significantly smaller combinatorial background.

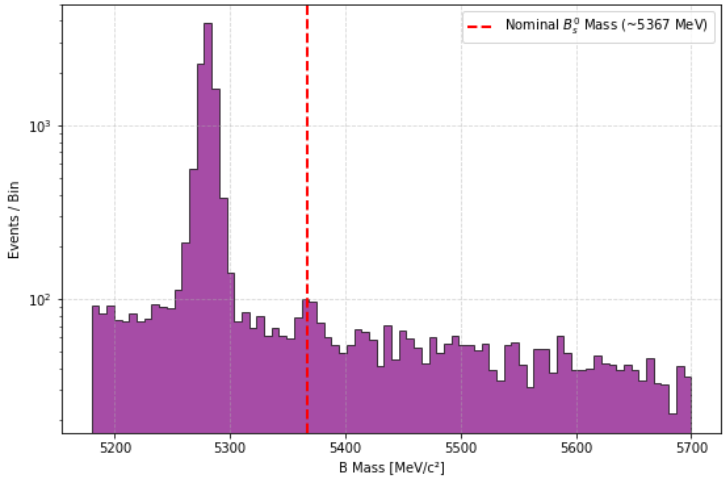


Figure 12: Invariant mass distribution of the real data candidates after applying the stringent BDT cut (> 0.975). Both the B^0 and B_s^0 peaks are clearly distinguishable from the residual background.

2.4 Estimation of the Number of Events

The signal yield from the surviving candidates was extracted using an Extended Unbinned Maximum Likelihood Fit. To accurately model the physical shape of the invariant mass peaks, including the asymmetric tails caused by energy loss, Crystal Ball probability density functions (PDFs) were utilized.

The fitting procedure was divided into steps to ensure stability: first, fits were performed on the MC samples to extract the true shape parameters (σ, α, n) of the B_s^0 signal and the B^0 control mode, shown in Fig 13 and 14. The parameters of these fits, summarized in Table 1 were then frozen.

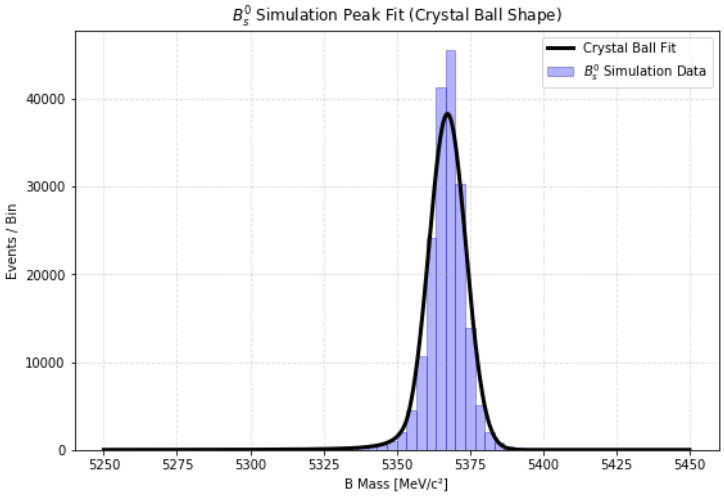


Figure 13: B_s^0 simulation mass distribution fitting with Crystal ball function

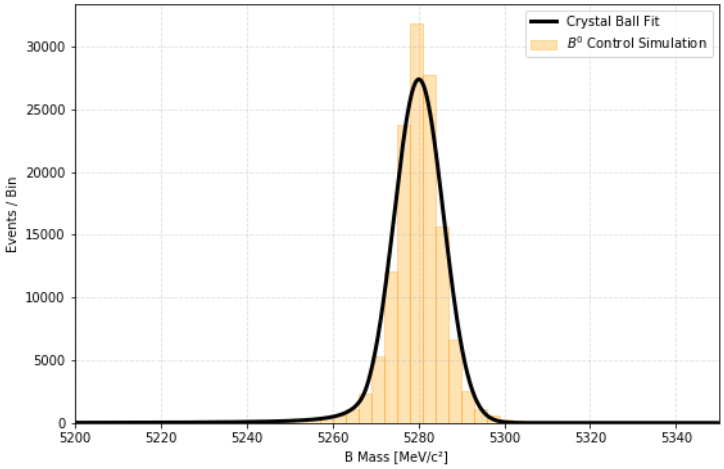


Figure 14: B^0 mass distribution fitting with Crystal ball function

Mode	σ	α	n
B_s^0 signal	6.31	2.08	2.35
B^0 control	5.73	2.11	1.98

Table 1: Fitting parameters of B_0^s and B^0 mass distributions

Finally, the extended unbinned fit was applied to the real data using a composite model consisting of the B_s^0 Crystal Ball, the B^0 Crystal Ball, and an exponential function to model the remaining combinatorial background. The results are shown in Fig 15.

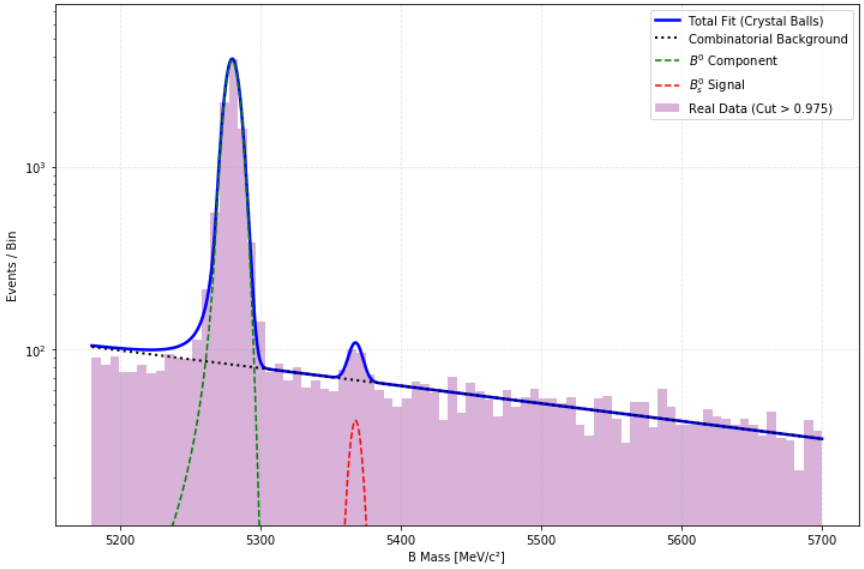


Figure 15: Invariant mass distribution of real data after apply optimal cut with combined fitting model (combinatorial background, B^0 and B_s^0 peak)

The number of signal and background candidates can be obtained from the integral over the signal window. These values are found to be $N_{sig} = 73.8$ and $N_{bkg} = 247.7$. We can also compute an estimated significance of the observation by,

$$m = \frac{N_{sig}}{\sqrt{N_{sig} + N_{bkg}}}$$

where N_{sig} and N_{bkg} are the observed numbers of signal and background events in the signal region, respectively. From the number of signal and background events, these correspond to the significance proxy $m = 4.12\sigma$.

3 Conclusions

A search for the rare $B_s^0 \to \psi(2S)K_S^0$ decay was performed. To isolate the signal from the combinatorial background, a multivariate classifier, specifically a Boosted Decision Tree (BDT), was trained using simulated signal events and real data. Following the classification, an extended maximum likelihood fit utilizing Crystal Ball functions was applied to the invariant mass distribution. This specific model was chosen to properly account for the kinematically similar

B^0 decay, which is not rejected by the classifier, combining the two meson peaks with an exponential background.

From the fit, the exact signal yield was extracted. By comparing the signal and background events within the signal region, a statistical significance proxy of 4.12σ was obtained. Exceeding the 3σ threshold, this result constitutes strong evidence for the observation of the $B_s^0 \to \psi(2S)K_S^0$ decay, although it falls short of the 5σ requirement necessary to claim a formal discovery.

For future improvements, additional kinematic features could be incorporated into the BDT to reject the background even more efficiently. Furthermore, an evaluation of the systematic uncertainties would be required to finalize the measurement.

References

[1] The LHCb Collaboration. The LHCb Detector at the LHC. *Journal of Instrumentation*, 3(08):S08005, Aug 2008.

[2] Tianqi Chen and Carlos Guestrin. XGBoost: A Scalable Tree Boosting System. New York, NY, USA, 2016. Association for Computing Machinery.

[3] Giovanni Punzi. Sensitivity of searches for new signals and its optimization, 2003.